System companies growing

The chart in Fig. 2 shows the percentage of the foundry customer revenue coming from the traditional integrated device manufacture (IDM) companies and from the new system companies. A decade ago, the systems companies were a trivial part of the foundry revenue. It has now increased and is likely to be about 21.3% next year, which represents a compounded annual growth rate of almost 27%. A lot of that has been in the areas such as Apple's iPhone but this trend is becoming common as these systems companies take more and more control over the architecture to deliver a better experience for their customers.

The four trends driving the digitisation strategies for these companies are sensors, edge computing, 5G/wireless communications, and cloud/data centres. Sensors provide the ability to digitalise the natural world and bring that into a compute infrastructure for new types of applications. Edge computing enables integration of devices into smaller footprints while allowing application of things like machine learning and artificial intelligence within those chips. 5G provides an infrastructure that enables pulling these devices together. Cloud and data centres allow taking this final set of data and deliver the end applications to the customers.

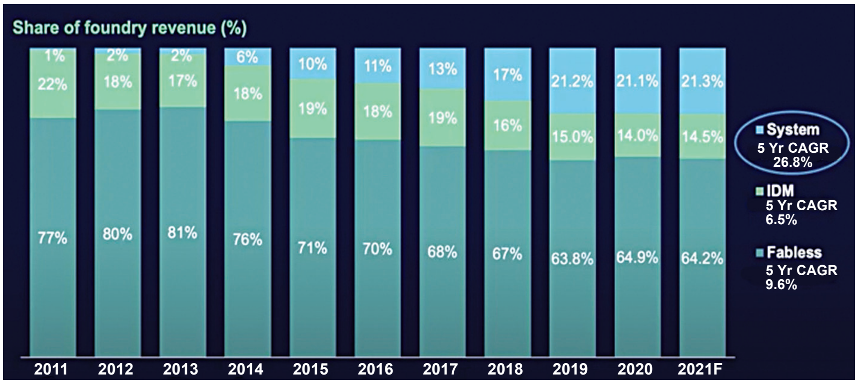

Fig. 2: Share of foundry revenue

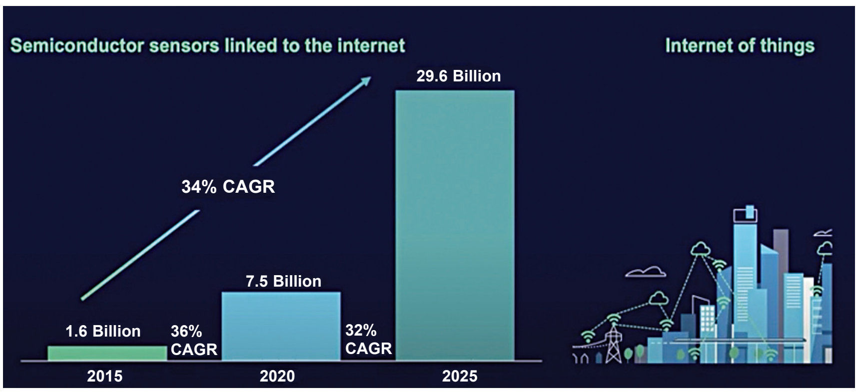

Fig. 3: Semiconductor sensors linked to the internet

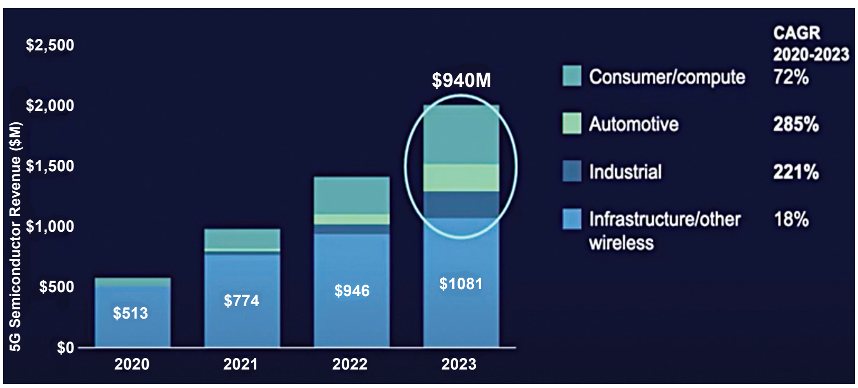

Fig. 4: Tremendous opportunity in 5G IoT applications

The chart in Fig. 3 compares the number of sensors linked to the internet in 2015 versus the number likely in 2025. As you can see, in 2015 there were about 1.6 billion sensors and by 2025 their number is projected to be almost 30 billion.

Fig. 4 shows a strong growth opportunity for 5G, which is expected to grow at about an 18% rate over the next few years. It also shows that consumer/compute, automotive, industrial, and infrastructure/other wireless applications are going to be an even larger driver of this opportunity as people start connecting those sensors and move that data into the data centres.

The chart in Fig. 5 shows a significant increase in growth and opportunities in data centres themselves. It shows a strong recovery from the dip in 2019 due to move to a virtualised work environment because of the pandemic. For the next five to six years the increase is likely at the rate of 11% CAGR in that large market, driving towards a 14% CAGR in later part of the decade.

Semiconductor market growth

The chart in Fig. 5 shows semiconductor growth moving forward. The chart on the left shows predictions that seem to be lining up these days for a trillion dollar semiconductor market around the year 2030. It can be seen that there is significant dif-